LIFELINE Iteration 1: Approach Note

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1. From story to numbers

General choices. Crate weight and value, road minutes and cable cost are uniform within the allowed ranges (the story gives no reason to prefer a shape). Sizes are drawn uniformly inside each tier range. Capacity is 10-70% of total crate weight, and on tier S it is chosen from the capacities where greedy-by-ratio is not optimal, so the 'obvious' loading fails.

1 Delta Flood (*raised roads go under water in waves*). A random tree plus extra roads gives a flat, near-tree delta with E chosen so the average degree is inside 2.2-3.2. 40% of villages (plus one) get penalty 2, 4 or 6 because flooded villages are slow to cross; the rest get 0 or 1. There are 3-5 waves. Food and water are 55% of crates (plus one) because *food and water matter most*.

2 Mountain Earthquake (*few roads, bursts after each aftershock*). Average degree 2.0-2.6 gives an almost-tree valley. Medicine and tools are 45% of crates (plus one). 70% of alarms are placed with a Gaussian around a few burst centres (about L/40 of them), the rest uniformly, and priorities are weighted high (mean about 7.3, accepted only above 6.05).

3 Coastal Cyclone (*short overlapping windows, noisy radio*). Shelters are 28% of locations (plus one) and there are 3-5 waves. Window length is raised step by step until the average strict overlap is at least 2.2, because windows are short but crowded. The log has 12% random symbols, and missing symbols are injected so all 38 appear (rule: at least 30).

4 Urban Flash Flood (*many alternatives, calls faster than anyone can answer*). Average degree 3.5-5.0. Alarms per tick are drawn in 1.52-1.78 on tiers M and L, so a queue must build up. Rare crates are 10% (plus one).

5 Landslide Corridor (*a single highway along a gorge*). The map is a path 0..V-1 with chords of span 2 or 3 and a maximum of 4 roads per location, so the farthest location is V-1 roads away. One bridge road is forced into wave 0 and target_0 is the end of the path behind it, so the union of waves cuts it off.

6 Outbreak Camp (*supplies spoil, beacons repeat*). Medicine and water are 55% of crates (plus one). More than half the crates get expiry 0-300. The log is six beacon messages repeated, and one symbol is forced to at least 27% of it (rule: 25%).

2. Planted objects and claims

Every planted object is built **first** and the rest of the data is built around it, so it is correct by construction. **Tour:** nearest-neighbour from site 0 (ties to the lowest site number), then 2-opt that only accepts shorter tours, so the planted tour is never longer than the nearest-neighbour tour; tour_length is computed from the final order. **Sensor cover:** k nodes are chosen first, then every edge is created with at least one endpoint in the cover (a path through the cover plus one edge from every other node gives N-1 edges, then extra cover-touching edges). **Safety rules:** the 0/1 assignment is chosen first; each rule takes 3 distinct switches with random signs, and if no literal is true one literal is flipped. **Rations:** random values, a hidden subset, and rations_target is its sum. **Radio:** patterns of length 2, 4 and 8 are cut from the log, so they occur; one longer pattern is random and checked to have count 0; counts use overlapping search. report_b is made from report_a by random edits and edit_bound is the exact edit distance from a DP. **Claims (tier S):** found by exhaustive search (subset DFS for cargo and windows, all orders for the tour, all subsets for the cover, recursion for the drum) and cross-checked in the generator against an independent DP or greedy; the generator stops if they disagree. **Variety rules:** rule 1 by choosing the capacity, rule 2 by swapping target_2 to a location whose fewest-roads route costs more, rule 3 by giving cases 20-45 the buckets (k-20) mod 3 (9 / 9 / 8 cases), rule 4 is checked by the validator. All are asserted at the end of generation.

3. The traps

46 Dock-order. 1,500-3,000 crates with value, expiry and serial sorted ascending and weight sorted descending, so value-per-kg is also ascending: all five fields are monotone. *Hostile:* a sort with a first- or last-element pivot degrades to $O(n^2)$, and any method that relies on random order loses its advantage. *Covered by V8* (monotone count of the five fields, at least 3).

47 Best-looking crate. 12-16 crates, capacity 20: one light lure crate (weight 1) with the best ratio, two crates of weight 10 and value 90-100, and fillers of weight 10-12. Greedy takes the lure and one heavy crate, then nothing else fits; the optimum is the two heavy crates. The generator asserts greedy is at most 60% of the optimum. *Hostile* to any ratio rule. *Covered by V8* (the validator recomputes greedy and the DP optimum).

48 Shortest looks cheapest. 1,500-3,000 locations. Each of the three targets has a gadget: a 2-road route through a penalty-6 junction (about 120 minutes) and a 5-9 road detour of 1-4 minute roads (under 40). A 900-1,000 road tail from the HQ makes the longest distance at least 500 roads, and the rest hang off it as bushes. *Hostile:* a BFS that stops at the fewest roads pays more than 1.5 times the cheapest route. *Covered by V8* (distance of at least 500, and one target with ratio at least 1.5).

49 Near-miss. 50,000-100,000 symbols, almost all one repeated symbol R with short noise patches. Patterns of length 8-40 are R repeated (length-1) times then a symbol that never occurs, so the first (length-1) symbols match at nearly

every position and the full pattern at none. *Hostile:* a naive matcher does about m comparisons at every position. *Covered by V8* (90% repetition of a block of at most 8, and at least four near-miss patterns).

4. What will be hard

These are expectations, to be checked by the Iteration 2 experiments.

1 Delta: *route planning*, because 40% high penalties make the cheapest route differ from the fewest-roads route and waves cut roads; *inventory*, because 55% food and water crates have similar ratios and day 3 doubles food, so greedy errors are large.

2 Mountain: *route planning*, because average degree about 2.3 means one closed road can cut a target off, so 'no route' must be handled; *alarm queue*, because bursts with mean priority above 6 create many equal-priority ties and a sudden backlog.

3 Cyclone: *rescue scheduling*, because each window overlaps at least 2 others, so highest-value-first blocks good windows; *route planning*, because 3-5 waves and 28% shelters give many reachable-but-costly targets.

4 Urban: *alarm queue*, because at least 1.5 alarms per tick with one release per tick grows the backlog to thousands (heap against list); *route planning*, because dense maps (degree at least 3.5) create many equal-cost routes, so the id tie rule must be exact.

5 Landslide: *route planning and the reachability check*, because one bridge cuts target_0 off and days 6-7 block every wave, so trips must be skipped; *the seven-day campaign*, because loading depends on a trip being possible.

6 Outbreak: *inventory*, because medicine and water are 55% of crates and day 5 triples medicine, so the optimal load changes between days; *rescue scheduling*, because the windows are unaffected by the supply story, so the value DP must stay exact while the daily unavailable-id rule removes windows.